

A 6.4 Tbps Optical Transmitter with Low-loss and High-Uniformity Inverse-Designed Multiplexer on a 300-mm CMOS Platform

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Abstract: We demonstrate a 6.4 Tbps transmitter by integrating 400 Gbps micro-ring modulators with an inverse-designed multi-dimensional multiplexer on a 300-mm CMOS platform, achieving low loss <1.5 dB, crosstalk <-18 dB, and ultra-low uniformity ($\sigma < 0.15$ dB). © 2026 The Author(s)

1. Introduction

Recently, inverse design has emerged as a powerful paradigm for developing ultra-compact and high-performance photonic integrated devices, enabling the discovery of non-intuitive device geometries. This methodology has been successfully applied to various multiplexing components critical for high-capacity optical interconnects, such as wavelength-division multiplexers [1] and mode-division multiplexers [2-4], with many demonstrations exhibiting compact footprints and broad bandwidths. However, the intricate patterns generated by inverse design have confined most devices to electron-beam lithography (EBL) rather than commercial foundry processes. The EBL-based fabrication approach prevents reliable large-scale manufacturing, restricting devices primarily to proof-of-concept demonstrations. Moreover, it hinders on-chip integration with active components, thereby obstructing the development of complex multi-functional photonic systems. Consequently, inverse-designed components remain largely at the stage of passive-device validation, with limited potential for practical application.

In this work, we implement a fabrication constraint and a calibration method to achieve high-performance and high-uniformity inverse-designed devices on a 300 mm CMOS platform, ultimately realizing a 6.4 Tbps multi-dimensional multiplexing transmitter. The fabricated multiplexer, occupying an ultra-compact footprint of $6 \times 6 \mu\text{m}^2$, supports four mode channels for two polarizations, exhibiting insertion loss below 1.5 dB and crosstalk under -18 dB for all channels. Notably, it achieves remarkable wafer-scale uniformity with a loss standard deviation below 0.15 dB. By integrating the highly uniform multiplexer with 400 Gbps micro-ring modulator (MRM) array, a large-capacity optical transmitter supporting 4 wavelength channels and 4 mode channels of two polarizations has been realized, which achieves a transmission capacity of 6.4 Tbps. This achievement represents a significant step towards the reliable volume production of high-performance, inversely designed photonic integrated circuits.

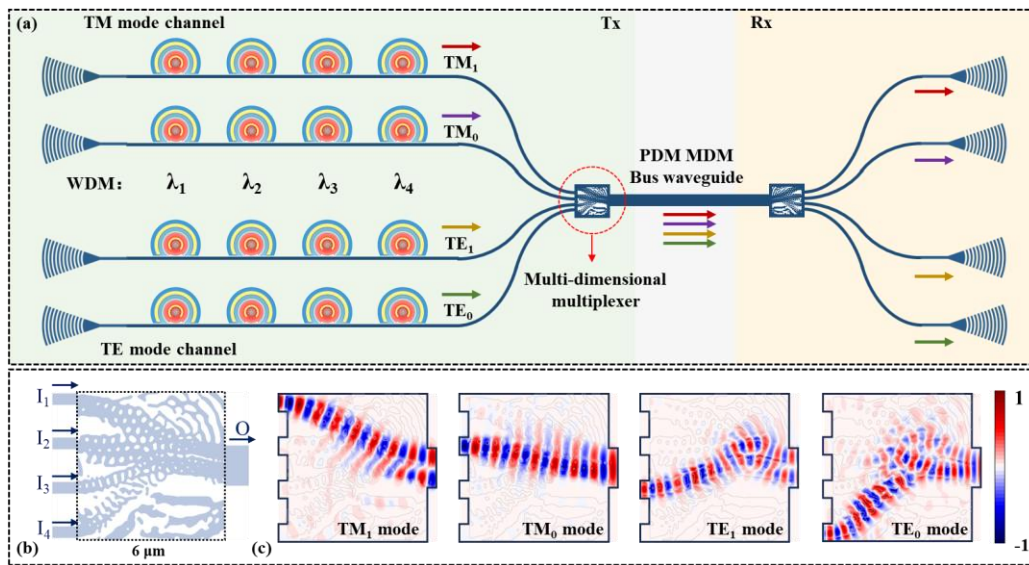


Fig. 1 (a) The schematic of proposed optical transmitter based on inverse designed multi-dimensional multiplexer. (b) Schematic of the proposed multiplexer. (c) The simulated mode field distributions for four mode transmission paths at 1550 nm.

2. Device design

Fig. 1(a) schematically illustrates the proposed multi-dimensional multiplexed optical transmitter. Four input fundamental modes are multiplexed by an inverse-designed multiplexer into a single multimode bus waveguide, where they are converted into distinct carriers— TE_1 , TE_0 , TM_1 , and TM_0 modes. The multiplexed signals then propagate along the bus waveguide before being demultiplexed and routed back into individual mode channels, where they are reconverted to fundamental modes. In each channel, an array of four micro-ring modulators modulates the fundamental-mode light. The micro-rings can operate at different wavelengths, enabling wavelength-division multiplexing (WDM). The designed (de)multiplexer exhibits a flat spectral response over the 1520–1580 nm wavelength range, allowing it to support multiple wavelength channels simultaneously. Fig. 1(b) presents the key multi-dimensional multiplexer optimized using adjoint method, which achieves multiplexing for two polarizations and two modes within a compact $6 \times 6 \mu m^2$ design area. Figure 1(c) shows the simulated mode field distributions for four mode channels at 1550 nm. The optimized multiplexer is capable of transmitting four fundamental modes from different input waveguides (I_1 – I_4) to the bus waveguide (O).

In our adjoint-based inverse design approach, a fabrication constraint method is implemented, which adds penalties to the objective function if the material density falls outside the range calculated based on filter radius and minimum feature size. Furthermore, a manufacturing calibration method (MCM) proposed in our previous work [2] that pre-process features of varying sizes generated by the adjoint method is also adopted.

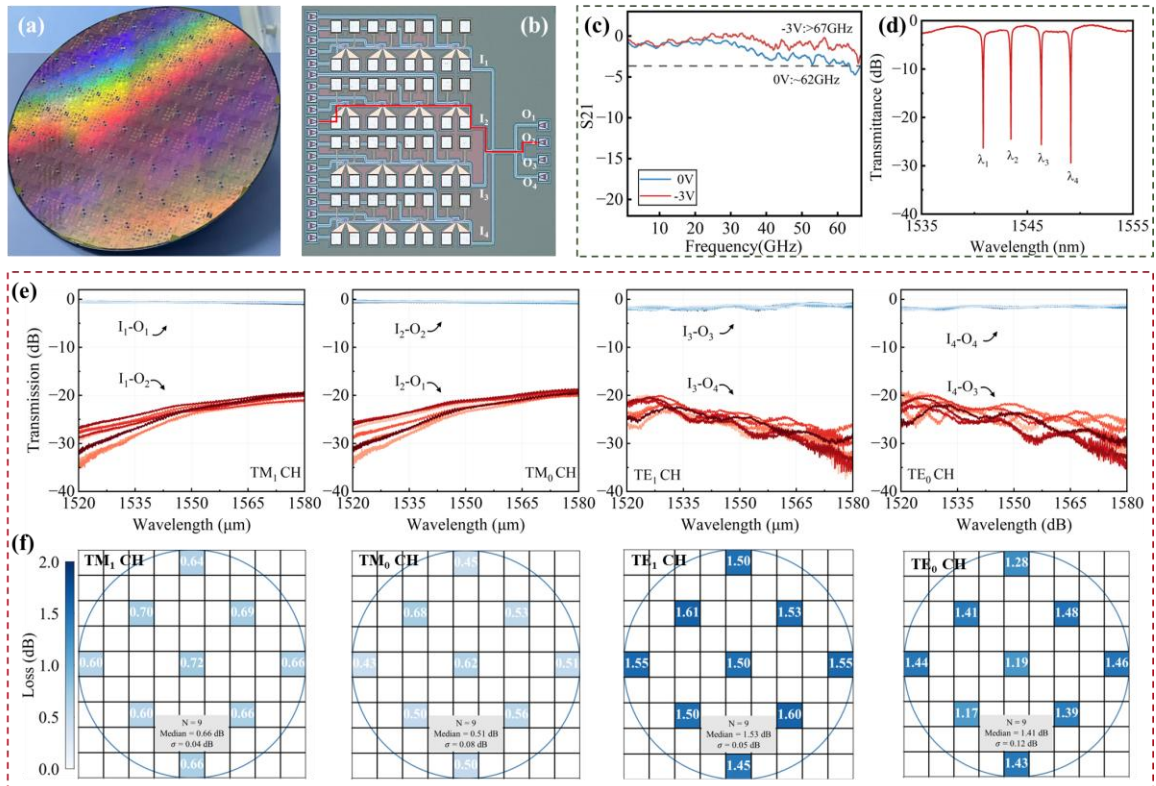


Fig. 2. (a) Wafer produced on 300 mm silicon photonic platform. (b) The microscope image of the fabricated optical transmitter circuit. (c) Electro-optical S21 response of micro-ring modulator. (d) Optical transmittance of the micro-ring modulator array. (e) Measured transmission spectra of four mode channels from 9 different chips across the wafer. (f) The insertion loss of the wafer level test results.

3. Experiment and measurement results

The inverse designed multiplexer is optimized based on the 220 nm silicon on insulator platform and fabricated on a 300 mm CMOS platform. The photograph of the fabricated 300-mm wafer and the microscope image of the optical transmitter are shown in Fig. 2(a) and Fig. 2(b) respectively. As shown in Fig. 2(c), the electro-optic (EO) 3 dB bandwidth of the MRM is measured to be 62 GHz at zero bias, and increases to beyond 67 GHz when a reverse bias of -3 V is applied. As shown in Fig. 2(d), the transmission spectra of the four-channel MRM array are presented. Through thermal tuning, channels 1 to 4 are evenly spaced by approximately 2 nm, enabling low-crosstalk data transmission and stable wavelength-division multiplexing.

To demonstrate the device performance and variability, we measured the insertion loss and crosstalk of four channels from 9 chips across the wafer and overlap the transmission spectra in Fig. 2(e). The fabricated multiplexer demonstrated high performance and uniformity across the wavelength range from 1520 nm to 1580 nm. The insertion loss for all four mode channels remains below 1.5 dB, with channel crosstalk under -18 dB, comparable to or even surpassing the performance of similar devices fabricated based on EBL. Moreover, the spectral responses of nine tested chips exhibit remarkable consistency, with their transmission spectra nearly overlapping across the entire band. To quantify this uniformity, we extracted the insertion loss at 1550 nm and calculated the standard deviation for each channel, as summarized in Fig. 2(f). The extremely low standard deviation of less than 0.15 dB for all four channels confirms excellent fabrication consistency and device reproducibility.

Finally, the high-speed data transmission performance across the four-channel system was experimentally characterized. The experimental setup is depicted in Fig. 3(a). The input optical signal was generated by a C-band tunable laser and the polarization was controlled using a polarization controller (PC). The RF signal is generated by a 256 GSa/s arbitrary wave generator. The RF signal is combined with -3 V DC bias by a bias-tee and loaded onto the MRM by an RF probe. The output laser was amplified by an erbium doped fiber amplifier (EDFA) and sampled by a digital communication analyzer to observe the eye diagrams. Figure 3(c) presents eye diagrams of four wavelength channels and four mode channels for 310 Gbps PAM4 and 400 Gbps PAM6. The transmitter dispersion eye closure quaternary (TDECQ) of all the channels for PAM4 signal transmission ranges from 2 dB to 2.5 dB. A total transmission rate of 6.4 Tbps can be achieved with a single-channel rate of 400 Gbps.

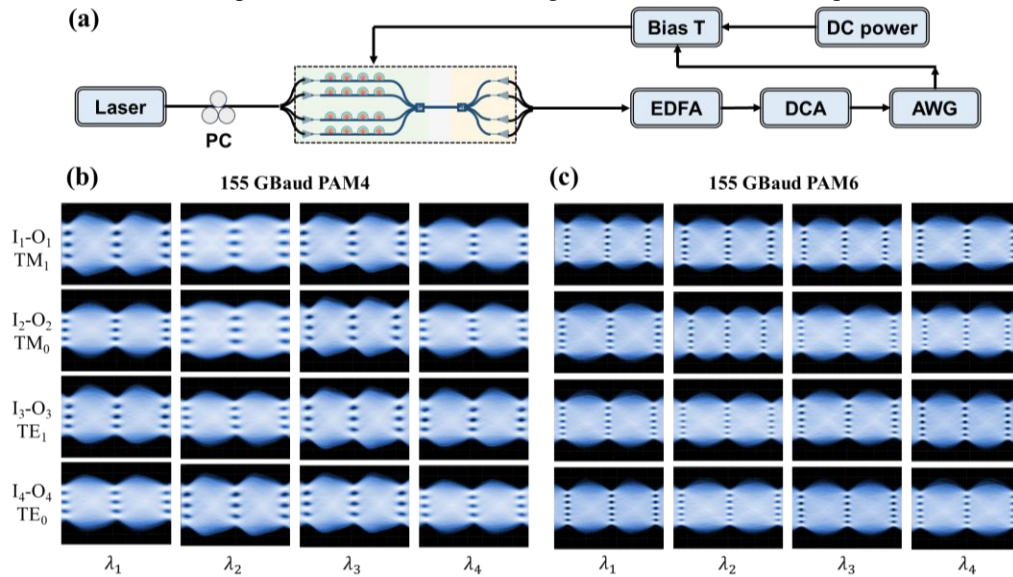


Fig. 3. The data transmission measurement. (a) Schematic of experimental setup. Measured optical eye diagrams for (b) 310 Gbps PAM4 (155 GBaud), (c) 400 Gbps PAM6 (155 GBaud) of four MDM channels with four wavelength carriers.

4. Conclusion

In conclusion, this work successfully demonstrates high-performance, high-uniformity inverse-designed photonic devices on a 300 nm CMOS platform, achieving a compact $6 \times 6 \mu\text{m}^2$ multiplexer with insertion loss < 1.5 dB, crosstalk < -18 dB, and remarkable wafer-scale uniformity ($\sigma < 0.15$ dB). Integrated with 400 Gbps micro-ring modulator arrays, a 6.4 Tbps multi-dimensional multiplexing transmitter is realized. This result paves the way for the reliable volume production of inverse-designed photonic integrated circuits, thereby enabling a new generation of multi-functional and ultra-compact systems through their compatibility with various active components.

5. References

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